```
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to
change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Main.java to edit this
template
 */

package launchgames;
import java.util.*;
import java.util.random.RandomGenerator;
class TicTacToe{
    static char [][]board;
    public TicTacToe(){
        board=new char[3][3];
        initBoard();
    }
    void initBoard(){
        for(int i=0;i<board.length;i++){
            for(int j=0;j<board[i].length;j++){
                board[i][j]=' ';
            }
        }
    }
  static  void display(){
        System.out.println("--------------");
         for(int i=0;i<board.length;i++){
             System.out.print(" | ");
            for(int j=0;j<board[i].length;j++){
               System.out.print(board[i][j] +" | ");
            }
             System.out.println();
             System.out.println("--------------");
        }
    }
   static void placeMark(int row,int coln,char mark){
        if(row>=0&&row<=2&&coln>=0&&coln<=2)
        board[row][coln]=mark;
        else System.out.println("Invalid Position so please enter the valid
position ");
    }
   static boolean checkColnWin(){
        for(int j=0;j<=2;j++){
            if(board[0][j]!='
'&&board[0][j]==board[1][j]&&board[1][j]==board[2][j]) return true;
        }
        return false;
    }
   static boolean checkRowWin(){
        for(int i=0;i<=2;i++){
            if(board[i][0]!='
```

```
'&&board[i][0]==board[i][1]&&board[i][1]==board[i][2]) return true;
        }
        return false;
    }
   static boolean checkDiagWin(){
        if((board[0][0]!=' '&&board[0][2]!='
')&&((board[0][0]==board[1][1]&&board[1][1]==board[2][2])||(board[0][2]==board[1][1
]&&board[1][1]==board[2][0])))
            return true;
        else return false;
    }
   static boolean CheckDraw(){
       for(int i=0;i<=2;i++){
           for(int j=0;j<=2;j++){
               if(board[i][j]==' ')return false;
           }
       }
       return true;
   }




}

abstract class Player{
     String name;
    char mark;
     abstract void MakeMove();
        boolean isValidMove(int row,int coln){
        if(row>=0&&row<=2&&coln>=0&&coln<=2){
            if(TicTacToe.board[row][coln]==' ')return true;
        }
        return false;
    }
}


class HumanPlayer extends Player{
//    String name;
//    char mark;
    public HumanPlayer(String name,char mark){
        this.name=name;
        this.mark=mark;
    }
    void MakeMove(){
        Scanner sc=new Scanner(System.in);
        int row;
        int coln;
        do{
```

```
            row=sc.nextInt();
            coln=sc.nextInt();
        }while(!isValidMove(row,coln) );
        TicTacToe.placeMark(row, coln, mark);
    }
//    boolean isValidMove(int row,int coln){
//        if(row>=0&&row<=2&&coln>=0&&coln<=2){
//            if(TicTacToe.board[row][coln]==' ')return true;
//        }
//        return false;
//    }
}
class AIPlayer extends Player{
//    String name;
//    char mark;
    public AIPlayer(String name,char mark){
        this.name=name;
        this.mark=mark;
    }
    void MakeMove(){
        Scanner sc=new Scanner(System.in);
        int row;
        int coln;
        do{
            Random r=new Random();
           row= r.nextInt(3);
           coln=r.nextInt(3);
        }while(!isValidMove(row,coln) );
        TicTacToe.placeMark(row, coln, mark);
    }
//    boolean isValidMove(int row,int coln){
//        if(row>=0&&row<=2&&coln>=0&&coln<=2){
//            if(TicTacToe.board[row][coln]==' ')return true;
//        }
//        return false;
//    }
}
```

```
public class LaunchGames {


    public static void main(String[] args) {
        TicTacToe t=new TicTacToe();
         HumanPlayer p1=new HumanPlayer("Appu",'X');
      // HumanPlayer p2=new HumanPlayer("Likhi",'O');
        AIPlayer p2=new AIPlayer("AILikhi",'O');
```

```
    //  HumanPlayer cp;
    Player cp;
                cp=p1;
                while(true){
                System.out.println(cp.name+" turn ");
                cp.MakeMove();
                TicTacToe.display();

if(TicTacToe.checkColnWin()||TicTacToe.checkRowWin()||TicTacToe.checkDiagWin()){
                System.out.println(cp.name+" has won ");
                break;
                }
                else if(TicTacToe.CheckDraw()){
                    System.out.println("game is draw");
                    break;
                }
                else
                    if(cp==p1)cp=p2;
                    else cp=p1;
                }

    }

}
```